

# Learning to Refer: How Scene Complexity Affects Emergent Communication in Neural Agents

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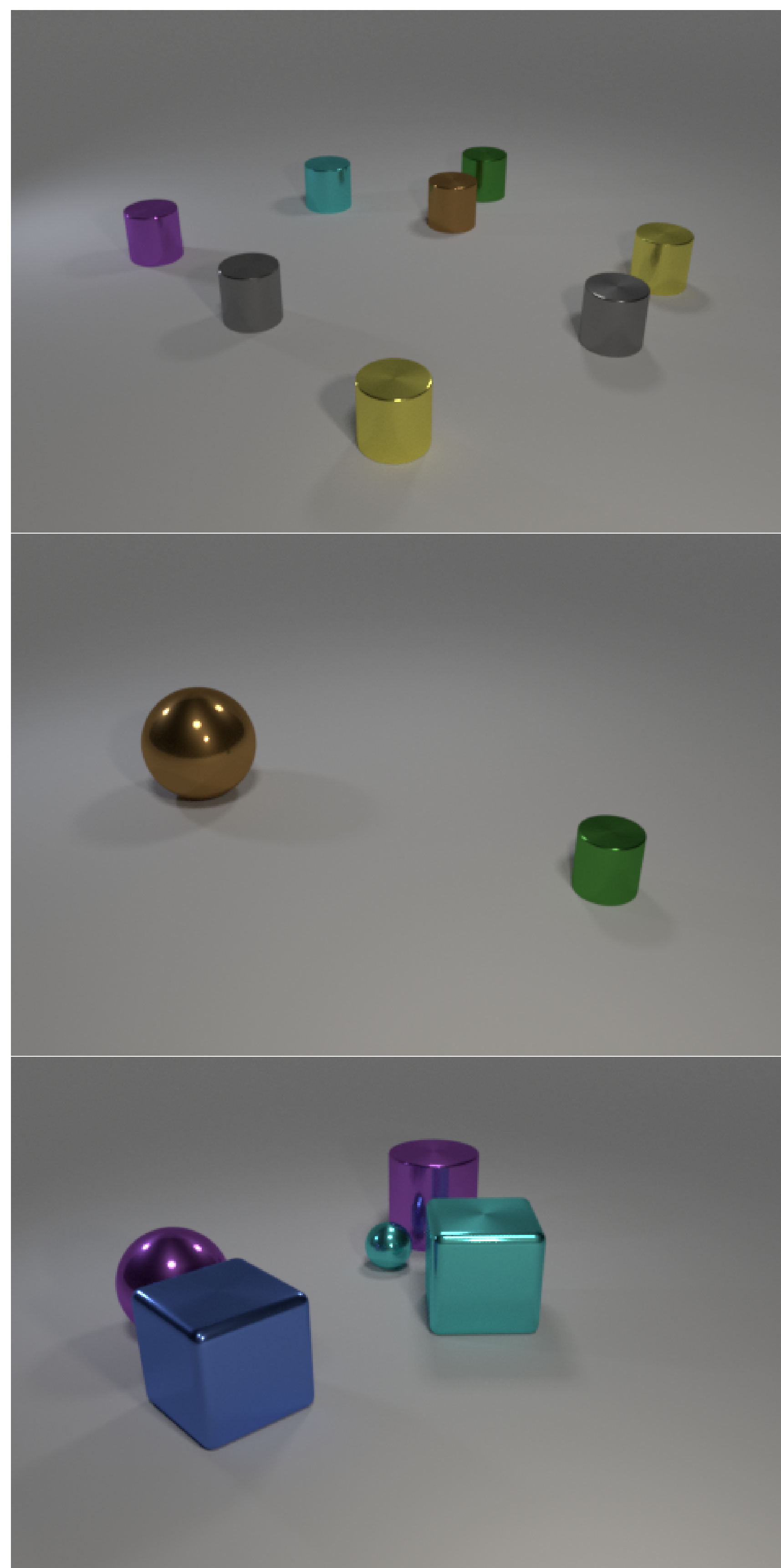
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## Introduction & Motivation

- How do cognitive systems bridge continuous sensory input with discrete symbols used in communication?
- Symbol grounding challenge: mapping visual features to meaningful linguistic descriptions
- We explore how neural agents learn this mapping through interactive language games
- Focus on referring expressions: describing objects uniquely in visual scenes



## Method

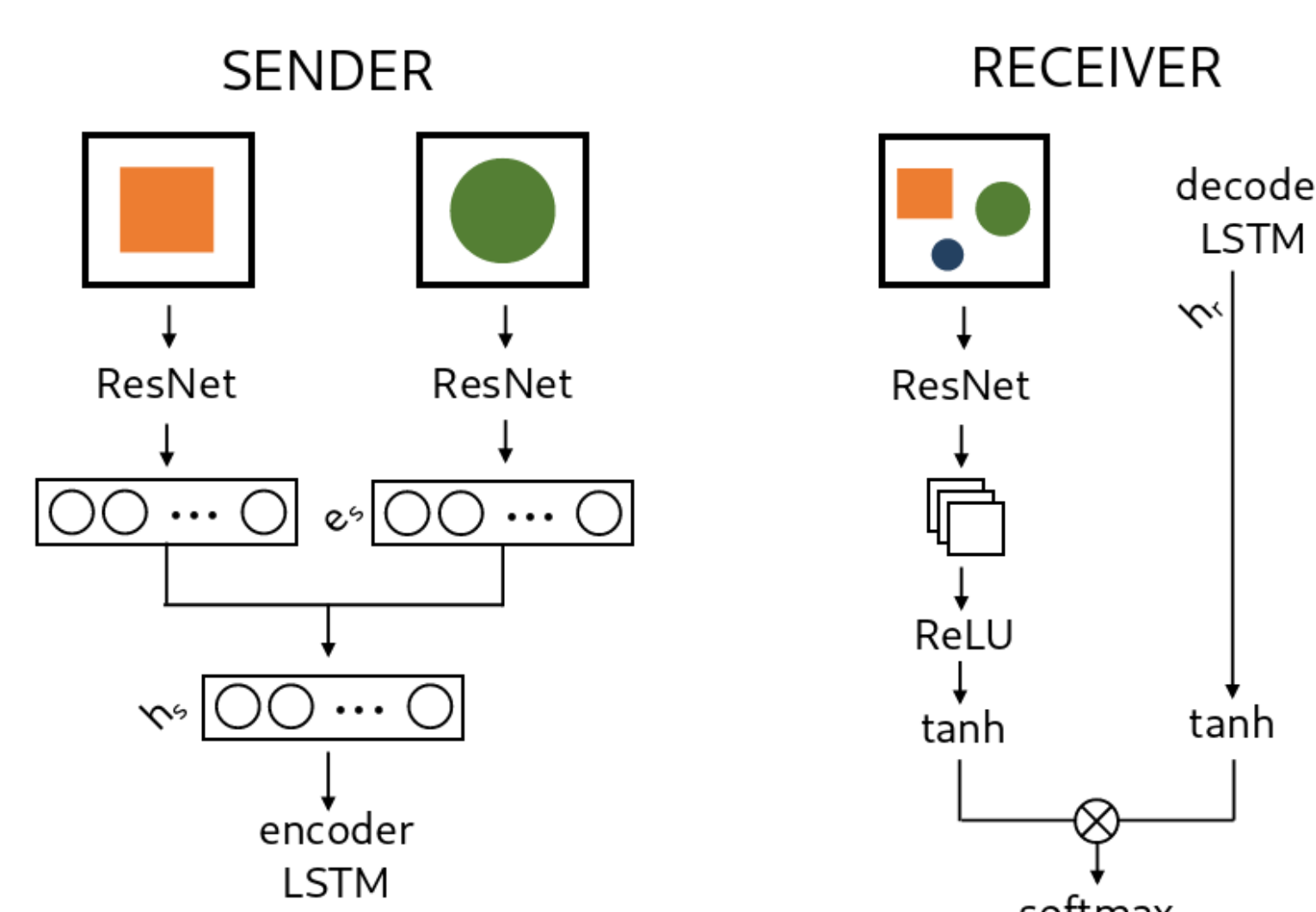
### Dataset Creation

We extend CLEVR to create three controlled datasets:

- **CLEVR color:** Target uniquely identified by color
- **Dale-2:** Target + 1 distractor
- **Dale-5:** Target + 4 distractors

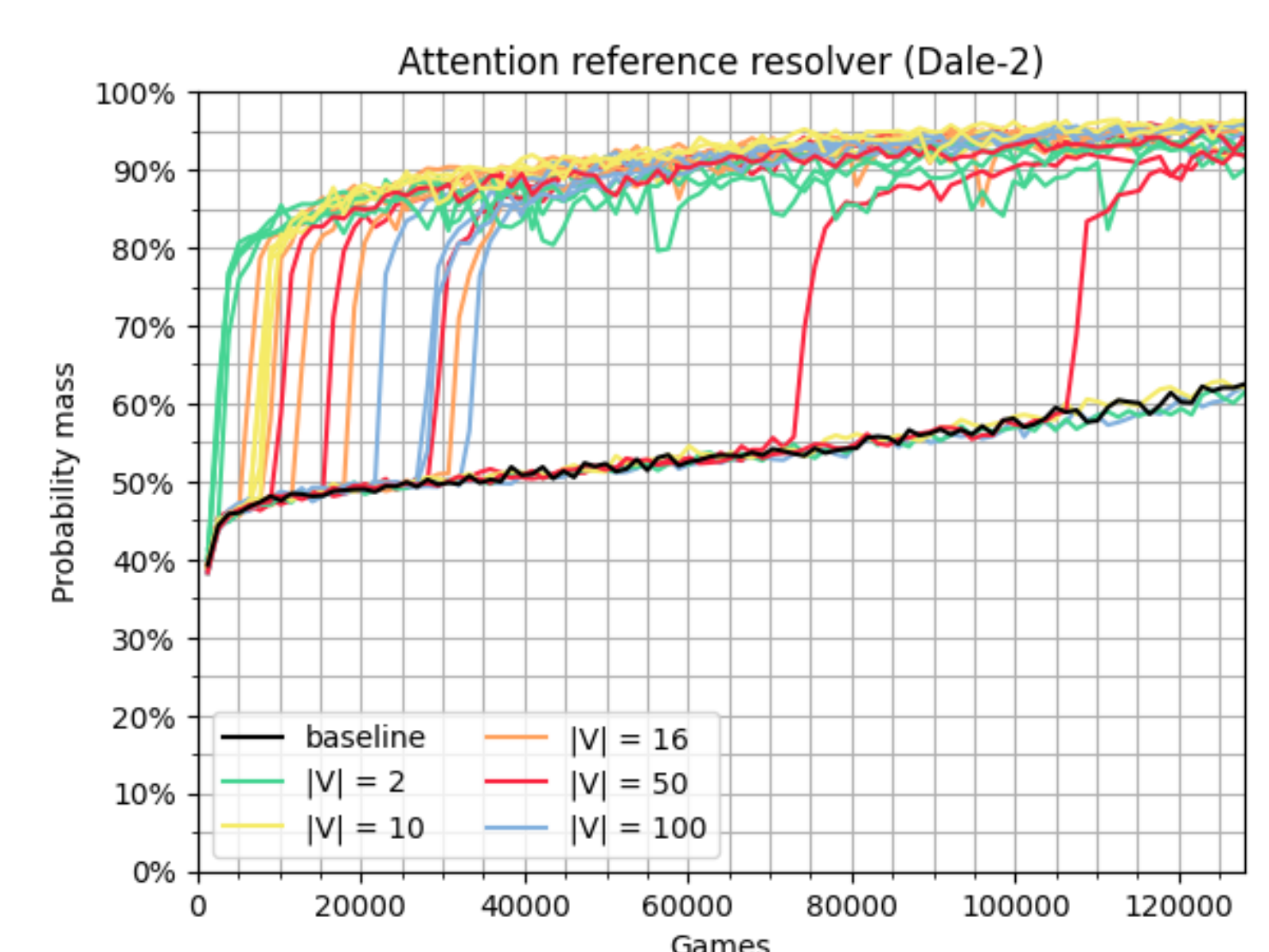
### Language Game Setup

- Two neural agents: Sender and Receiver
- Sender sees target object, generates message
- Receiver sees full scene, interprets message
- Goal: Identify target object location
- Learning through interaction feedback



## Results

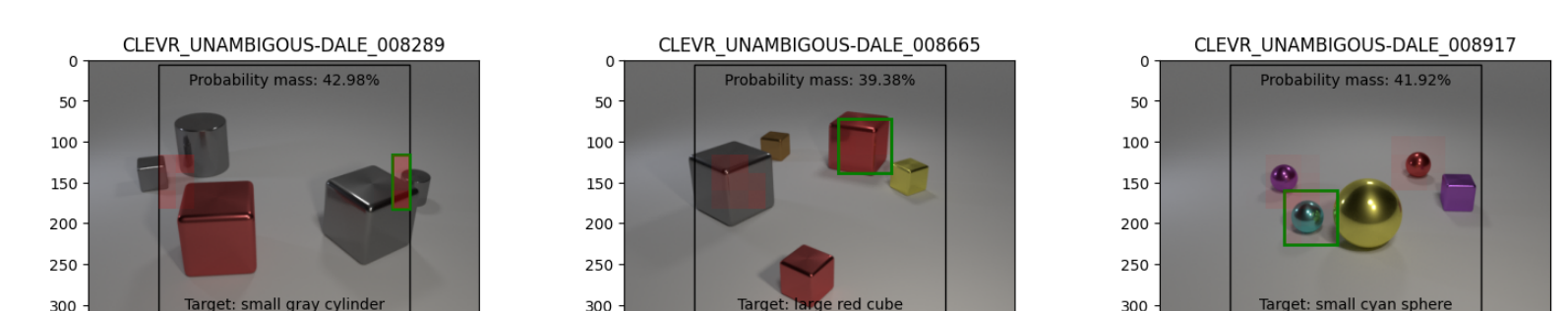
### Language Game Performance



### Key Findings:

- Medium-sized vocabularies (10-16 symbols) perform best
- Message length of 2-4 symbols optimal
- Performance decreases with scene complexity:
  - Dale-2: Up to 96% accuracy
  - Dale-5: Up to 84% accuracy
  - CLEVR color: Up to 68% accuracy

### Error Analysis



### Common Error Patterns:

- Confusion between objects sharing multiple attributes
- Split attention between similar objects
- Edge cases with partially visible objects

## Conclusions

- Scene complexity significantly impacts emergent communication

- Shape attributes are easiest to learn and communicate
- Size attributes prove most challenging
- Human-like pragmatic referring requires explicit modeling
- Results inform design of language-vision systems

## References

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